

A 75-year-old man presents with an acute dissection of the descending thoracic aorta. His blood pressure measures 190/110 mmHg. Justify your plan for his preoperative blood pressure management. (30%) Describe the potential complications of thoracic endovascular aortic repair (TEVAR) and how you would mitigate them. (70%)

A) Blood pressure management is critical in acute dissection to prevent the worsening of dissection and also aortic rupture. This acute control needs to be balanced with maintaining perfusion to end-organs specifically the gut, kidneys and spinal cord

Monitoring:

- 5 lead ECG, IA line insertion and CVC insertion, KIV cardiac output monitor
- 5 lead ecg good to pick out myocardial ischemic changes, IA line will allow for invasive blood pressure monitoring and closer titration of BP drugs and CVC will allow us to deliver vasodilatory medications centrally.
- Cardiac output monitor will guide fluid resus and vasodilatory/vasopressor medications titration
- Risks of IA line insertion, CVC insertion needs to be weighed against.
- Aim for SBP 100-120 and Hr 60-80. Balance between maintaining perfusion and reducing likelihood of further dissection

Drugs:

1. Labetalol bolus and infusion. Labetalol boluses of 10-20mg followed by a 0.5-2mg/min infusion titrated to effect.
 - Both a beta and plus alpha blocker. Effective in bringing down blood pressure, reducing afterload and reducing cardiac contractility/Heart rate
 - Beta blockers should be given before vasodilators as reflex catecholamine release due to vasodilation may increase LV contraction
2. Nitrates. GTN infusion 10-200mcg/min. This is mainly a venodilator with some vasodilator effects. Usually used as 2nd agent to control
3. Calcium channel blockers can be given if beta-blockers are contra-indicated
4. Treat the presence of any pain with IV fentanyl 25mcg or morphine boluses of 1-2mg

Exclude other causes of increase in blood pressure like urinary retention, anxiety.,

B)

Patient complications:

Aortic dissection causing ischemia to spinal cord, gut, kidneys. This can cause abdominal compartment syndrome, colitis and AKI

> optimise perfusion by keeping MAP higher >80, 100% Fio2, correct any anemia (aim Hb >9)

> insertion of lumbar drain to reduce CSF volume and improve spinal cord perfusion (SCFP = MAP – CSFP)

> Decompressive laparotomy needed for abdominal compartment syndrome

Acute coronary syndromes: acute increase in afterload can cause increase myocardial demand and thus offset the balance of myocardial demand vs supply leading to intra-op cardiac events.

> to reduce intra-op BP with agents such as labetalol infusion, GTN infusion so that SVR can be reduced, HR can be reduced and thus reduce myocardial demand

Increased risks of stroke: if left subclavian artery also involved in dissection

> If done under MAC, to monitor neurological status closely and titrate sedation

> if done under GA monitor cerebral perfusion with cerebral oximetry and optimise perfusion as mentioned above. To use only short acting GA medications so that post-op neurology can be checked immediately

Patient discomfort and uncooperative patient affecting successful deployment of stent. If procedure decided to be done under MAC

- > careful patient selection. If not able to cooperate, to do procedure under GA.
- > To ensure adequate local analgesia given at groin, IDC inserted (full bladder from contrast induced diuresis)

Anaesthesia complications:

Loss of airway during MAC sedation:

- > Close monitoring of patient if done under MAC with etco2 monitoring and careful titration of sedative agents.

Hemodynamic surges intra-op causing aortic rupture. This can happen during airway manipulation, intubation (if case done under GA) or pain during femoral cannulation (when done under MAC)

- > Adequate analgesia with TCI remifentanyl infusion to be started before any airway procedures.
- > adequate local analgesia given by surgeons at insertion site if done under MAC. To be supplemented with small boluses of IV fentanyl or titrated IV premedex infusion for systemic analgesia

Surgical complications:

- Contrast induced nephropathy: endovascular repair uses large amounts of contrast to guide treatment and stent placement.
 - > To monitor urine output closely and to ensure adequate fluid resuscitation. To avoid other nephrotoxic drugs.

- aortic perforation: Inadvertent aortic injury can cause rupture causing severe blood loss and need to convert to open procedure

- > Ensure level 1 infusers, airway equipment on standby in the event of rupture for rapid fluid/blood infusions and conversion to GA if not already done
- > may need to reverse heparin with protamine to control bleeding

Remote location with radiation:

- > Ensure all staff are wearing lead suits and to ensure senior staff and equipment on standby at remote location

Lower limb ischemia and arterial occlusion as a result of thrombus

- > distal pulses should be checked at the end of the op and if recognised, urgent embolectomy needs to be performed